MITHIL K GOWDA

+91-8431196506 mithil.k.g.10@gmail.com LinkedIn GitHub Portfolio

Professional Summary

Cybersecurity enthusiast with hands-on experience in secure software development, vulnerability assessment, threat modeling, and backend engineering. Project Intern at ISRO–LEOS with experience developing secure applications using Python, Flask, and MySQL. First author of internationally accepted cybersecurity research, seeking graduate opportunities in Security Engineering, SOC, Threat Intelligence, and Cloud Security.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru 2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Cybersecurity: Threat Detection, Threat Intelligence, Vulnerability Assessment, Threat Modeling, Network Security, Web Security, Secure Software Development, OWASP Top 10, Security Architecture

Tools: Kali Linux, Wireshark, Nmap, Burp Suite, OWASP ZAP

Programming: Python, Java, JavaScript, SQL, HTML, CSS

Frameworks: Flask, Node.js, REST APIs

Databases: MySQL, MongoDB

Cloud: Google Cloud Platform (GCP)

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend modules using Python, Flask, REST APIs, and MySQL to support analytics-driven engineering applications and internal operational workflows.
- Implemented secure user authentication, database integration, and structured application workflows to enhance system reliability, controlled access, and data integrity.
- Built automated data processing and visualization pipelines to support engineering analysis, operational reporting, and data-driven decision-making.
- Independently developed application features deployed for internal organizational use, ensuring scalability, maintainability, and operational efficiency.

Selected Cybersecurity Projects

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of a cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good.*
- Designed a multi-agent cyber defense architecture integrating intelligent threat detection, adaptive deception, autonomous response, and continuous learning to strengthen resilience against sophisticated cyber attacks.
- Engineered the Detect–Mislead–Neutralize–Learn security lifecycle by combining XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, and distributed threat intelligence.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed a secure digital carbon trading ecosystem integrating Digital Carbon Passports, intelligent governance, fraud detection, trust scoring, and secure transaction validation.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, authentication mechanisms, and secure transaction verification.

C^3T-STAVP Cryptographic Framework (Ongoing Research)

- Designing a hybrid cybersecurity framework integrating Character Chaffing & Transposition Technology (C^3T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching Zero-Knowledge Proofs (ZKP), semantic obfuscation, post-quantum cryptography, privacy-preserving architectures, and secure transaction verification mechanisms.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM (Coursera)
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL
- Google Cloud Computing – NPTEL
- IBM Front-End Developer Specialization – IBM (Coursera)

Achievements

- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized cybersecurity research focused on AI-driven cyber defense, autonomous threat detection, and intelligent security architectures.
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.